

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Java, C++, TypeScript, JavaScript, SQL, Bash

AI Agents & LLMs

AI agents, tool use, MCP (basics), prompt engineering, RAG, LLM evaluation, OpenAI / Groq APIs, Ollama, HuggingFace Transformers

ML & Data

PyTorch, NumPy, pandas, scikit-learn, classification / regression, data cleaning & preprocessing, statistics (CIs, hypothesis tests), matplotlib, generative models

Data Engineering

Apache Spark / PySpark, Spark UI, ETL pipelines, large-scale data processing, SQL analytics

Backend & Cloud

FastAPI, Node.js, REST APIs, Docker, AWS (S3, KMS), GCP, Azure, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.
- Joined a **GitHub Copilot** productivity pilot, routing tech-debt PRs, code migrations, and security-finding triage through **AI-assisted workflows** and maintaining a conventions file to keep AI-generated code aligned with team patterns.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies from conversation history and user input; iterated on prompt design and response-quality heuristics for a real customer-facing experience while building 20+ Node.js + Fastify REST APIs over PostgreSQL.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that converts open-ended topic queries into structured entity tables using typed retrieval planning, page rerank, deterministic extraction with LLM fallback, output verification, and **cell-level provenance** for every result.
- Hardened service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

AutoEval - Retrieval Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Built an evaluation-improvement agent that proposes new queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates back to Git with report cards for an auditable, reproducible evaluation workflow.

Spark ETL - Large-Scale Data Cleaning & Preprocessing

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, SQL, Python | 1.4B-row dataset

- Cleaned, transformed, and **preprocessed a 1.4B-row dataset** with a Spark ETL pipeline (schema-aware reads, partitioning, column pruning, derived features), cutting end-to-end runtime **25%** and profiling stage timings via Spark UI.

Deep Learning Models & Optimization

github.com/Deva-1903/UMASS_CICS_689

PyTorch, JAX, NumPy | UMass CS 689 Advanced ML

- Trained and benchmarked **5 deep architectures** (MLP, VGG-style CNN, ResNet) for **image classification** on CIFAR-10 across 3 optimizers, and implemented a reverse-mode **autodiff engine from scratch** (validated against JAX) plus a diffusion model in JAX.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.